

Gustavo Henrick

Software Engineer | Node.js · TypeScript · Python

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SUMMARY

I build transactional systems, where money, inventory, and data integrity cannot be wrong. My work sits on the sale path of retail point-of-sale (POS) software, split between test automation and moving the product from a legacy stack to cloud microservices. What sets me apart is verification discipline: I do not accept a conclusion without a positive control. I find the class of defect that passes the test and reaches the customer.

EXPERIENCE

Decision Group, on assignment at Diebold Nixdorf

Software Engineer · Sep 2025 to present · São Paulo, Brazil

Test automation and architecture migration on the point-of-sale system of a major Brazilian retailer.

Test automation, built end to end

- Defined test scope upstream of the automation: wrote test plans and scripts, set acceptance criteria, and decided what belongs in the regression suite.
- Designed and built the full automation engine in Python, from the execution runner to the payment, login, and validation modules, covering **67 automated flows** across sales, bill payments, and reporting.
- Built a **custom DSL with an editor** on an element-mapping layer that gave the team autonomy: QA engineers author new tests without writing code and without going through me.
- Changed how QA spends the cycle: the regression battery, previously run by hand on every change, now runs in the background, and the team reviews results instead of executing them.
- Wired the suite to **Selenium, SSH, SQLite, and SQL queries**, so results are verified in the database and not only on screen.
- Implemented **data integrity and tax validations** that supported analysis of real data inconsistencies.

Migration to microservices

- Refactored and migrated a critical sale flow written more than ten years ago and running in production, moving it from the legacy stack to cloud microservices in **Node 22, TypeScript, and Fastify** with hexagonal architecture, without changing the screen the cashier uses.
- Wrote and maintained **242 tests**, with on-screen validation through Chrome DevTools Protocol and screenshot evidence per run.

GTSofthub, own product

Software Engineer · Nov 2025 to present

Multi-tenant sales platform for small retailers: POS, inventory, storefront, and a WhatsApp AI agent that handles customers and closes orders. Currently in QA, with acceptance testing completed at a real store. **1,104 commits, sole author.**

- Enforced tenant isolation **in the database**, through Row-Level Security under a restricted role, not application-level filtering. Suites run under that same role, not as superuser, which would bypass the rule and exercise a mode production never uses.
- Built the platform's test suite: **1,404 cases across 156 files**, with 17 integration suites running against **real Postgres and Redis** as a blocking gate in the CI pipeline.
- Found and fixed a systemic false positive: **13 of 17 integration suites reported success without executing anything**, because a missing environment variable broke initialization and the error was swallowed. The suite was green and protected no one.
- Put a **deterministic guard** on the model output: every monetary value the AI states is checked against the database before it reaches the customer, and any claim it cannot verify is dropped. Validated by a **63-scenario** harness that never calls a live LLM.
- Covered overselling and inventory race conditions with transactional integrity tests, because concurrent sales are where that class of bug shows up.

PROJECTS

DevFinder Pro · semantic search over GitHub profiles

github.com/eoGuuga/devfinder · devfinder-frontend · live and public

- Modeled search with **embeddings (Sentence Transformers)** over a **Pinecone** vector database, matching on meaning instead of keyword overlap.
- Split the system into two services: a **Python/FastAPI** API in **Docker** on **Hugging Face Spaces**, and a **React/Vite** front end on **Render**.

EDUCATION

B.S. in Computer Science · Universidade Cruzeiro do Sul, São Paulo, Brazil · 2024 to 2027, in progress

AWS Microcredentials · Agentic AI Demonstrated and Serverless Demonstrated · hands-on exams · 2026, valid through 2027

Full-Stack Software Engineering Program · Rocketseat, a Brazilian software engineering school · 181 hours · 2025

Cyber Threat Management · Cisco Networking Academy · 2025

SKILLS

Languages: TypeScript, JavaScript (Node 22), Python, SQL, Bash, PowerShell

Backend and web: NestJS, Fastify, TypeORM, hexagonal architecture, REST APIs, Next.js

Data: PostgreSQL (Row-Level Security, roles, migrations), Redis, SQLite

Infrastructure: Docker, GitHub Actions, Nginx, Linux VPS, versioned deployments

Quality: test planning, acceptance criteria, and regression scope definition, Jest, Vitest, Supertest, Selenium, integration testing against real dependencies, concurrency and transactional integrity testing, Chrome DevTools Protocol

Applied AI: embeddings and vector search, deterministic verification of model output, scenario-based evaluation

Ways of working: Scrum, Jira, code review, Git, Bitbucket